Slow Fermentation Improves Crumb Structure in Sourdough Bread

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Abstract
We study how fermentation time changes the crumb of sourdough bread. Loaves proofed for eighteen hours at a cool temperature had larger and more even air cells than loaves proofed for four hours at room temperature. Slow fermentation also raised acidity, which tasters preferred. The effect held across three flours and two starter cultures.

1 Introduction

Sourdough bread rises because wild yeast and lactic acid bacteria ferment the sugars in flour. The yeast makes carbon dioxide, which inflates the dough, and the bacteria make acids, which give the bread its sour taste. Bakers have long believed that a slow, cool fermentation gives a better loaf, but the claim is rarely measured.

In this paper we measure it. We baked one hundred and twenty loaves under controlled conditions and scanned each crumb with a flatbed scanner. Image analysis gave the size and spacing of every air cell. We then asked a panel of tasters to rate each loaf without knowing how it was made.

2 Methods

Each loaf used five hundred grams of flour, three hundred and fifty grams of water, ten grams of salt and one hundred grams of active starter. Half the loaves were proofed for four hours at twenty-four degrees. The other half were proofed for eighteen hours at six degrees in a refrigerator. All loaves were baked in the same oven for forty minutes.

We used three flours: a white bread flour, a whole wheat flour and a rye blend. Two starter cultures were kept on a fixed feeding schedule for a month before the study began. The crumb was scanned at six hundred dots per inch after the loaves had cooled overnight.

3 Results

Slow fermentation produced larger air cells in every flour. The median cell area rose by a third for the white flour and by a quarter for the whole wheat flour. Cell size was also more even: the spread of cell areas fell in every case. The rye blend changed the least, which we attribute to its weaker gluten network.

Tasters preferred the slowly fermented loaves in seventy percent of paired comparisons. They most often described them as tangier and more complex. Acidity measurements agreed: the slow loaves were about half a pH unit more acidic.

4 Conclusion

A long, cool fermentation gives sourdough a more open and even crumb and a taste that most people prefer. Home bakers can get the benefit with no extra work by proofing their dough overnight in the refrigerator. Future work should test more flours and longer proofing times.

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